

AI STUDYBUDDY API

PROJECT DESIGN PHASES

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SYSTEM DESIGN OVERVIEW

1. Introduction

The **Project Design Phase** is the stage where the requirements and analysis of the AI StudyBuddy system are transformed into a technical design.

The design defines how different components of the application communicate with each other and how information flows through the system.

The major components of AI StudyBuddy are:

- User Interface
- REST API
- Authentication Module
- AI Service
- Database
- Study Management Modules
- Progress Tracking Module

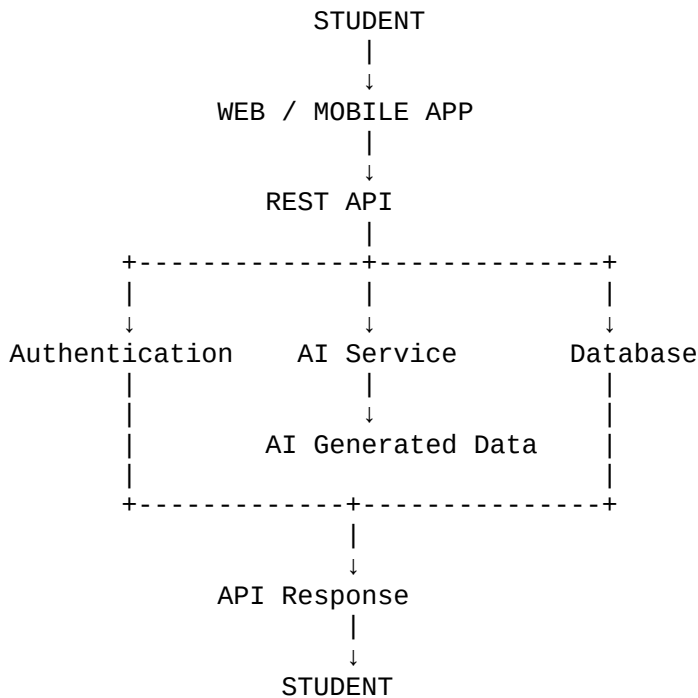
2. Design Objectives

The main objectives of the design phase are:

1. Create a simple and understandable system architecture.
 2. Define communication between frontend and backend.
 3. Design REST API endpoints.
 4. Design the database structure.
 5. Provide secure authentication.
 6. Make the system scalable.
 7. Make the API easy to maintain.
 8. Support future AI-based features.
-

3. Overall System Architecture

The proposed architecture is:



4. Architecture Components

Frontend

The frontend provides the interface through which students interact with StudyBuddy.

It may contain:

- Login page
- Dashboard
- AI Chat page
- Notes page
- Quiz page
- Flashcards page
- Progress page

Backend API

The backend receives requests from the frontend and processes them.

It handles:

- Authentication
- Validation
- Business logic
- AI communication
- Database operations
- Error handling

AI Service

The AI service processes educational requests and generates:

- Answers
- Notes
- Summaries
- Questions
- Flashcards

Database

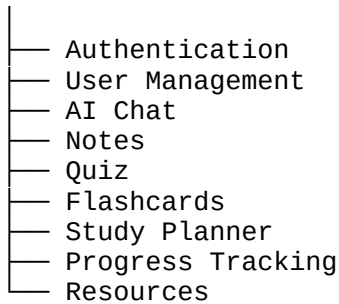
The database stores user accounts, notes, quizzes, flashcards, and learning progress.

MODULE AND API DESIGN

1. Modular Design

The AI StudyBuddy backend can be divided into separate modules.

AI STUDYBUDDY API



Modular design makes the system easier to develop and maintain.

2. Authentication Module

The authentication module manages student accounts.

Register

POST /api/auth/register

Login

POST /api/auth/login

Profile

GET /api/auth/profile

Logout

POST /api/auth/logout

3. AI Chat Module

The AI Chat module allows students to ask educational questions.

POST /api/chat

Example request:

```
{  
  "message": "Explain machine learning in simple words"  
}
```

Example response:

```
{  
  "success": true,  
  "response": "Machine learning is a method..."  
}
```

4. Notes Module

Generate Notes

POST /api/notes/generate

Get Notes

GET /api/notes

Create Note

POST /api/notes

Update Note

PUT /api/notes/{id}

Delete Note

DELETE /api/notes/{id}

5. Quiz Module

Generate Quiz

POST /api/quizzes

Example:

```
{
  "topic": "Python",
  "number_of_questions": 5,
  "difficulty": "easy"
}
```

Get Quiz

GET /api/quizzes/{id}

Submit Quiz

POST /api/quizzes/{id}/submit

6. Flashcard Module

POST /api/flashcards
GET /api/flashcards
DELETE /api/flashcards/{id}

Flashcards contain:

```
{
  "question": "What is RAM?",
  "answer": "Random Access Memory"
}
```

DATABASE AND DATA DESIGN

1. Database Design

The database stores information required by the StudyBuddy application.

A relational database can contain tables such as:

Users
Notes
Quizzes
Questions
Flashcards
Progress
Chat_History

2. Users Table

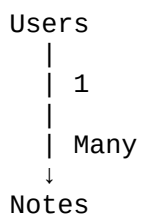
Field	Type	Description
user_id	INT	Primary key

Field	Type	Description
name	VARCHAR	Student name
email	VARCHAR	User email
password_hash	VARCHAR	Secure password hash
course	VARCHAR	Course
year	INT	Academic year
created_at	DATETIME	Account creation date

3. Notes Table

Field	Type	Description
note_id	INT	Primary key
user_id	INT	User reference
title	VARCHAR	Note title
content	TEXT	Note content
created_at	DATETIME	Creation date

Relationship:



One user can create multiple notes.

4. Quiz Table

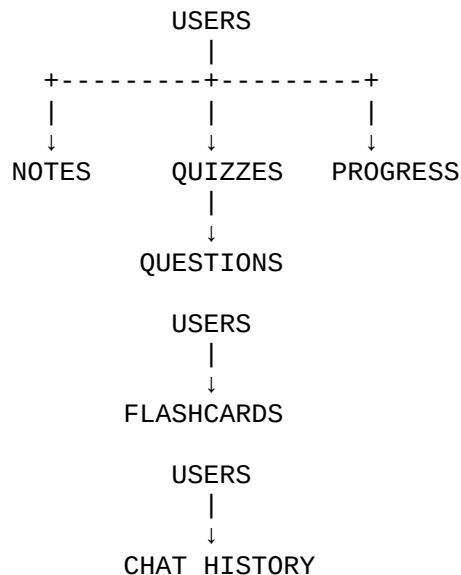
Field	Type	Description
quiz_id	INT	Primary key
user_id	INT	User reference
topic	VARCHAR	Quiz topic
difficulty	VARCHAR	Difficulty
score	INT	Quiz score
created_at	DATETIME	Creation date

5. Progress Table

Field	Type	Description
progress_id	INT	Primary key
user_id	INT	User reference
activity	VARCHAR	Activity type
score	INT	Score

Field	Type	Description
completed_at	DATETIME	Completion time

6. Entity Relationship Structure

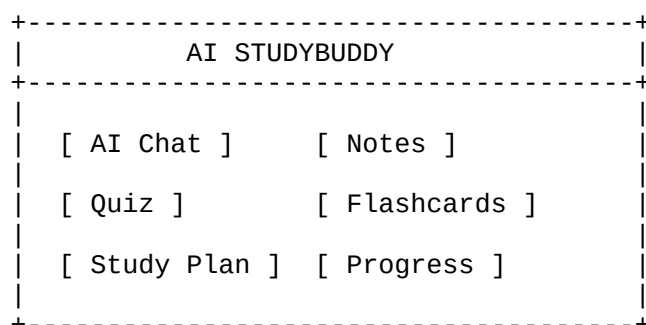


The `user_id` connects user-specific learning data.

PAGE 4 – UI, SECURITY AND API FLOW DESIGN

1. User Interface Design

The application dashboard can be designed as:



The design should be simple so students can quickly access learning features.

2. AI Chat Screen

AI CHAT	
Student:	What is cloud computing?
AI:	Cloud computing is the delivery of computing services over the internet.
Type your question...	[Send]

3. Quiz Screen

QUIZ
Question 1
What is Python?
☐ Operating System ☐ Programming Language ☐ Database ☐ Browser
[Next]

4. Security Design

Security should be considered throughout the API design.

Authentication

Protected endpoints should require an authentication token.

Authorization: Bearer <token>

Password Security

Passwords should never be stored as plain text. They should be securely hashed before storage.

Input Validation

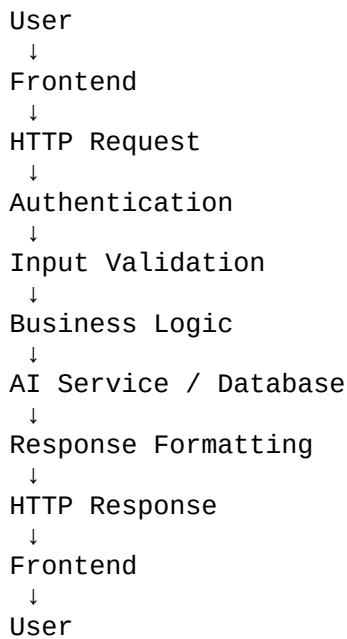
The API should validate:

- Email format
- Password requirements
- Required fields
- Data types
- Request size

API Key Protection

AI service API keys should be stored on the backend and should not be exposed to frontend users.

5. API Request Flow



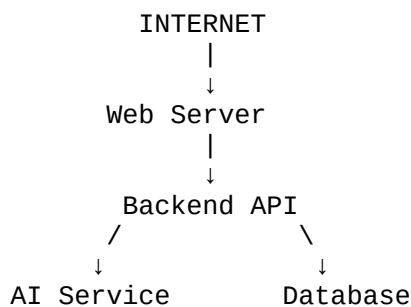
This structure separates the user interface from backend processing.

DEPLOYMENT, TESTING AND DESIGN

CONCLUSION

1. Deployment Design

The AI StudyBuddy system can be deployed using the following structure:



The frontend can be hosted separately from the backend API.

Example:

Frontend

↓
https://studybuddy.example
Backend
↓
https://api.studybuddy.example

2. Development Environment

A possible technology stack is:

Component	Technology
Frontend	HTML, CSS, JavaScript / React
Backend	Python / Node.js / PHP
API	REST
Database	MySQL / PostgreSQL
AI Service	AI Model API
Authentication	JWT / Session-based authentication
Server	Cloud / Local Server

The exact technologies can be changed according to project requirements.

3. API Testing Design

Each API endpoint should be tested independently.

Example

POST /api/auth/login

Test cases:

Test Case	Expected Result
Correct email and password	Login successful
Incorrect password	Authentication error
Empty email	Validation error
Invalid email	Validation error
Missing password	Validation error

Quiz APIs should similarly be tested with valid and invalid topics, question counts, and difficulty values.

4. Error Handling Design

The API should provide standardized error responses.

```
{  
  "success": false,
```

```
    "message": "Invalid authentication token",
    "error_code": "AUTH_001"
}
```

Common HTTP status codes:

Code	Meaning
200	Successful request
201	Resource created
400	Bad request
401	Unauthorized
403	Forbidden
404	Not found
429	Too many requests
500	Server error

5. Future Design Enhancements

The system can later support:

- Voice-based AI interaction
 - Multilingual learning
 - PDF/document analysis
 - Image-based questions
 - Personalized learning recommendations
 - Advanced learning analytics
 - Mobile applications
 - Offline study materials
 - Gamification
 - Integration with educational platforms
-

6. Conclusion

The **Project Design Phase of AI StudyBuddy API** converts the requirements and analysis into a technical blueprint for implementation.

The design defines the **system architecture, API modules, database structure, user interface, authentication mechanism, security controls, request flow, testing approach, and deployment structure**.

The complete design can be summarized as:

```
Requirements
  ↓
System Architecture
  ↓
Module Design
  ↓
API Design
```

↓
Database Design
↓
UI Design
↓
Security Design
↓
Testing & Deployment
↓
AI STUDYBUDDY

This design provides the foundation for the next development stages, including **coding, API implementation, database integration, frontend development, testing, and final deployment.**